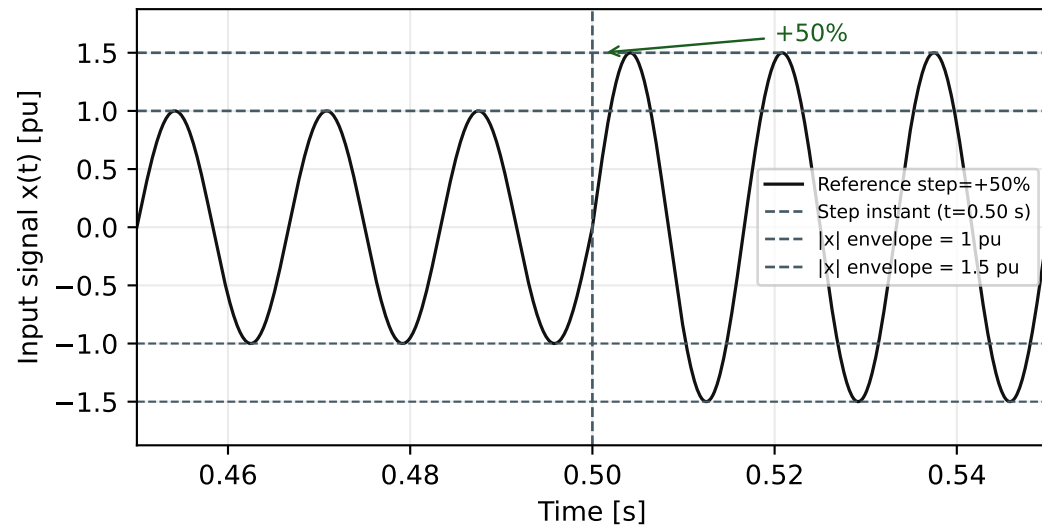


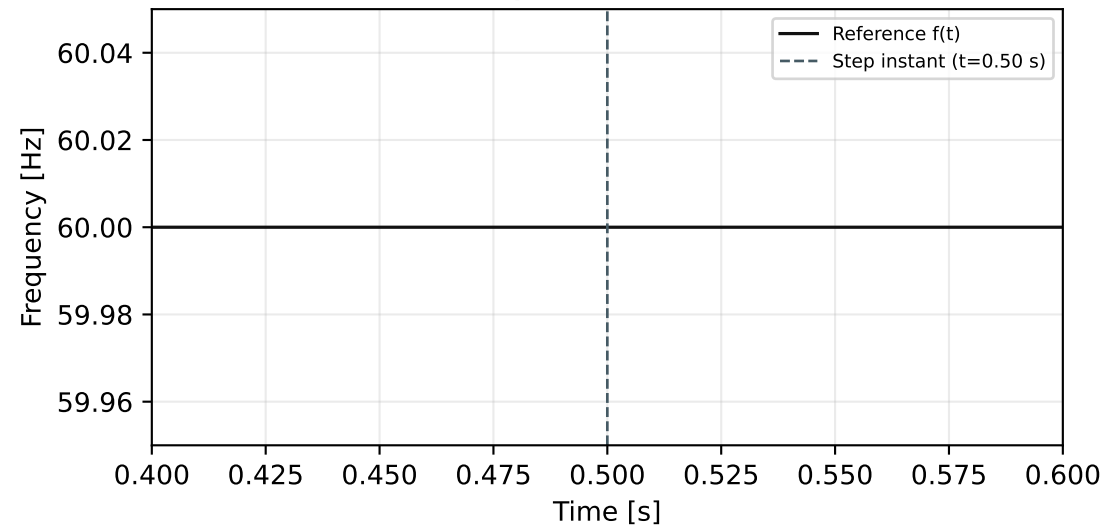
RMSE: by estimator family

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify AM-to-FM coupling, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits.

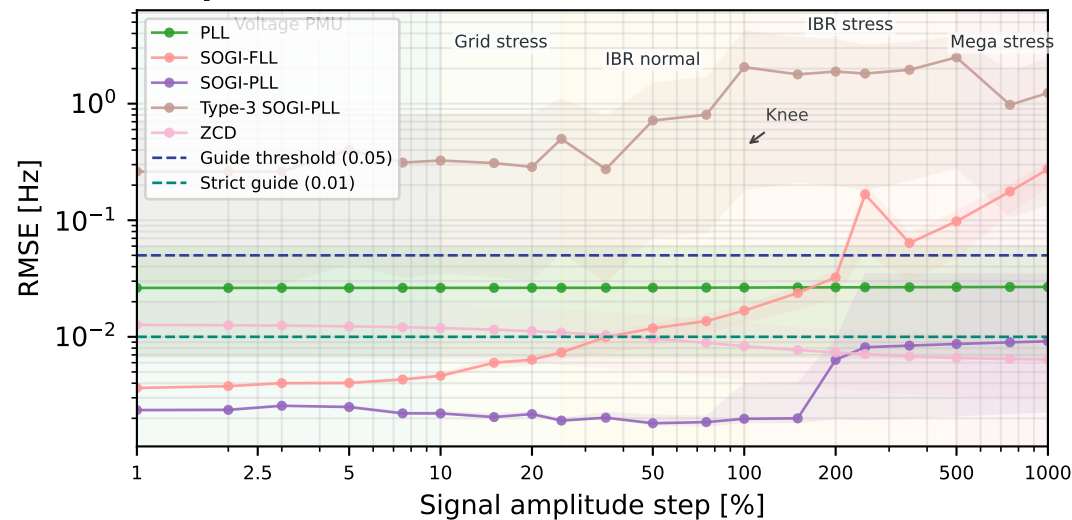
Reference Amplitude Step



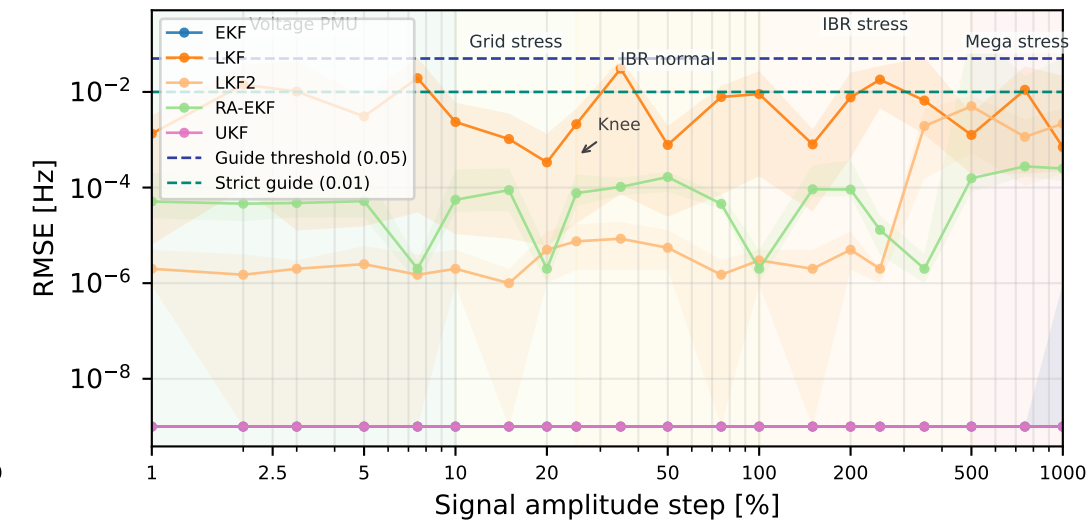
Reference Frequency $f(t)$



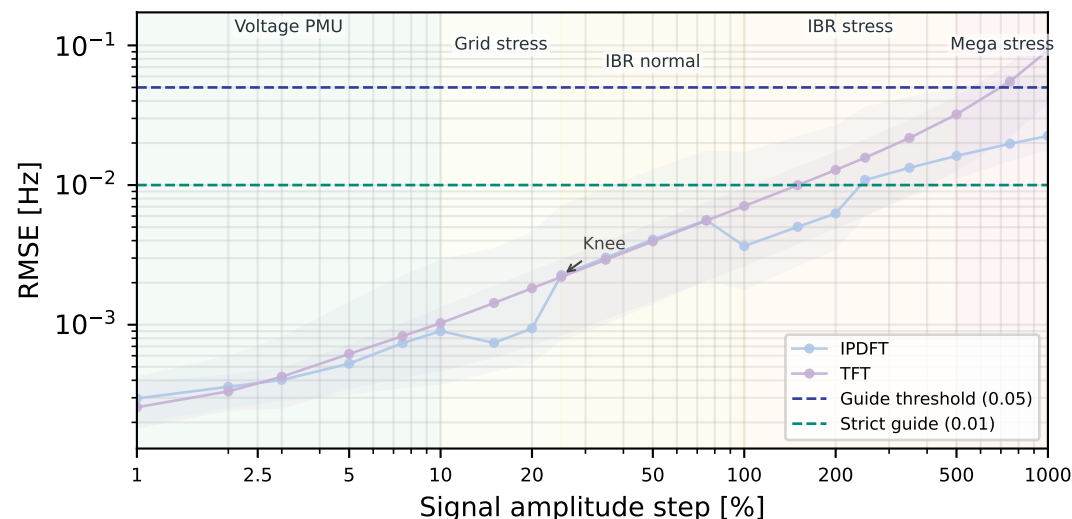
Loop-based



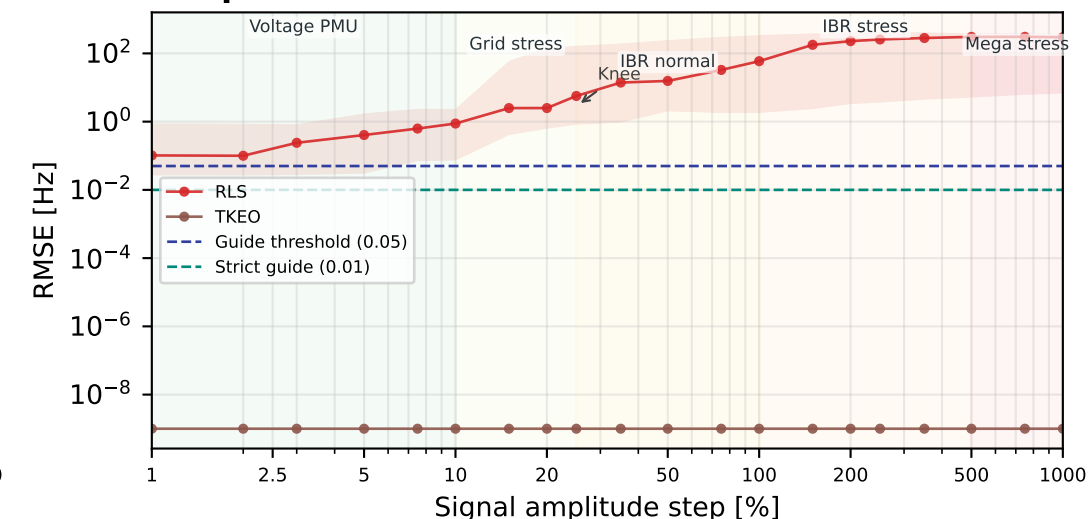
Model-based



Window-based

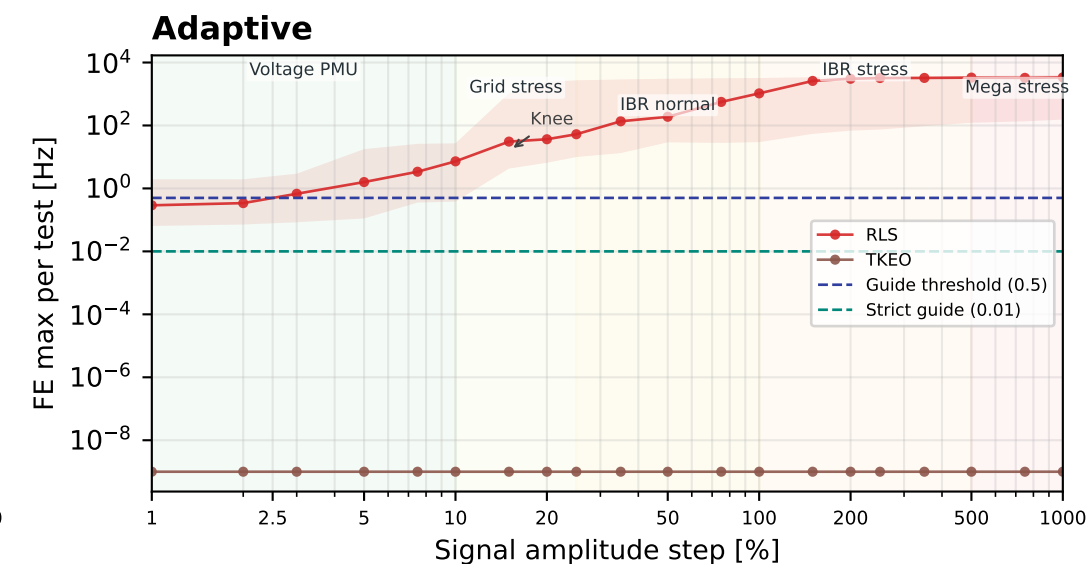
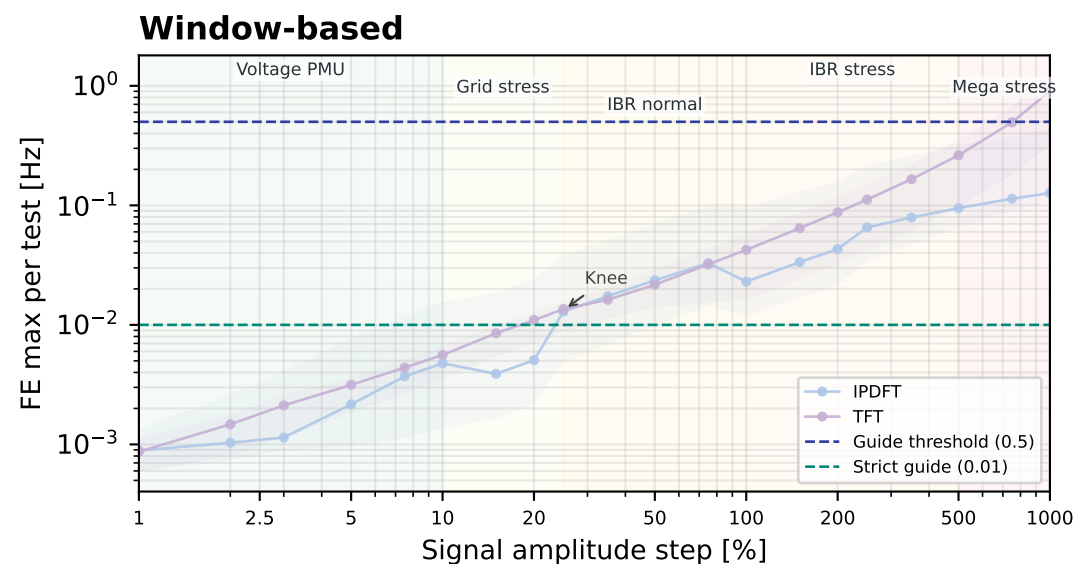
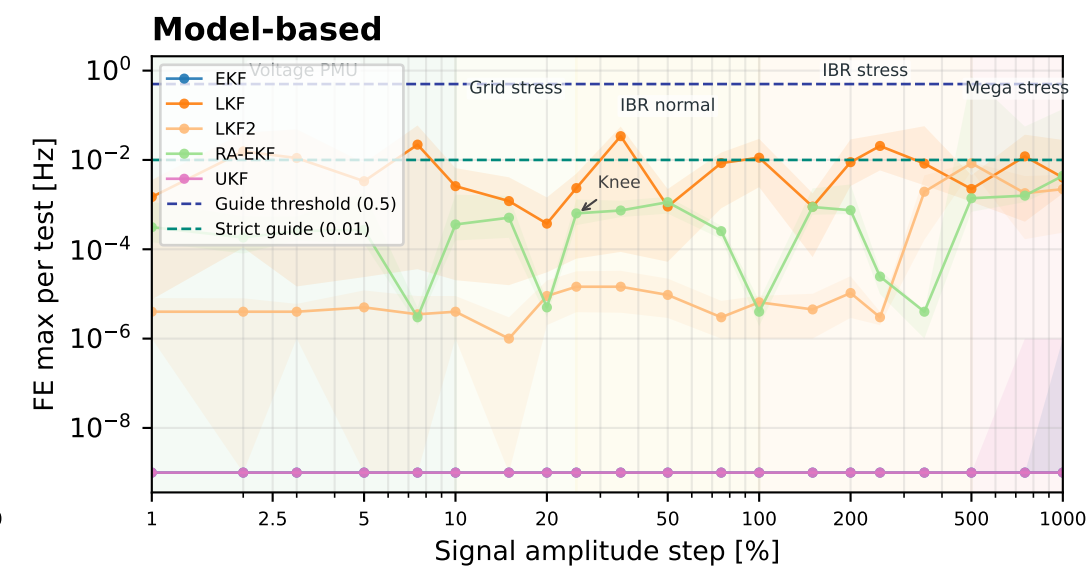
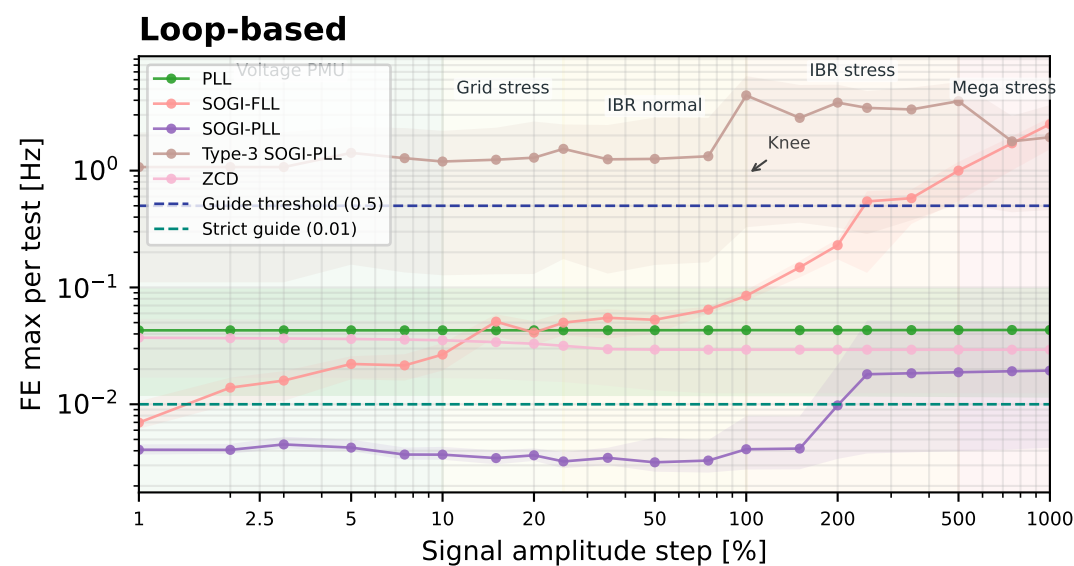
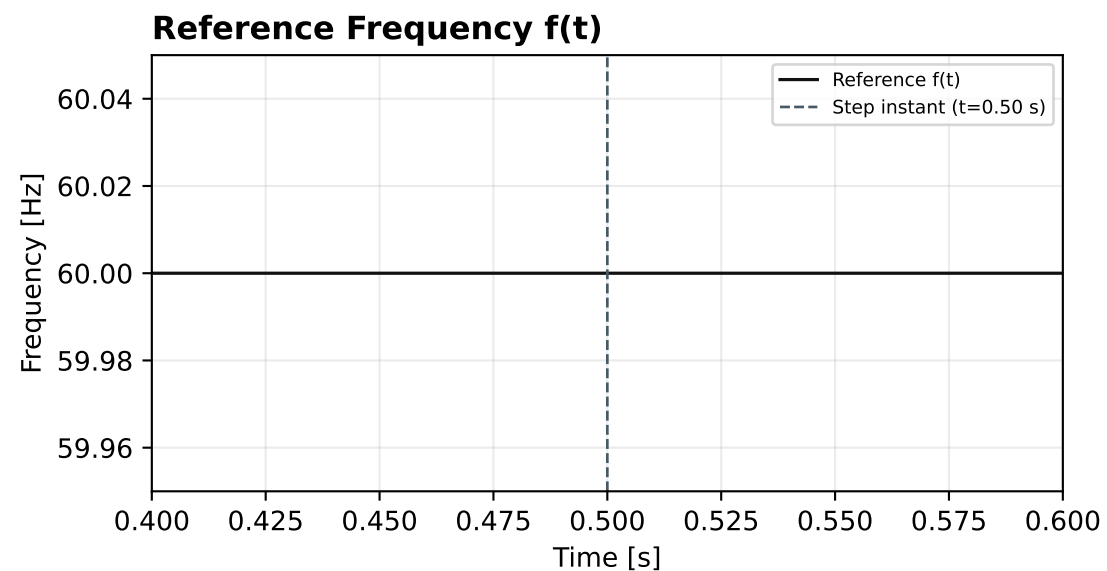
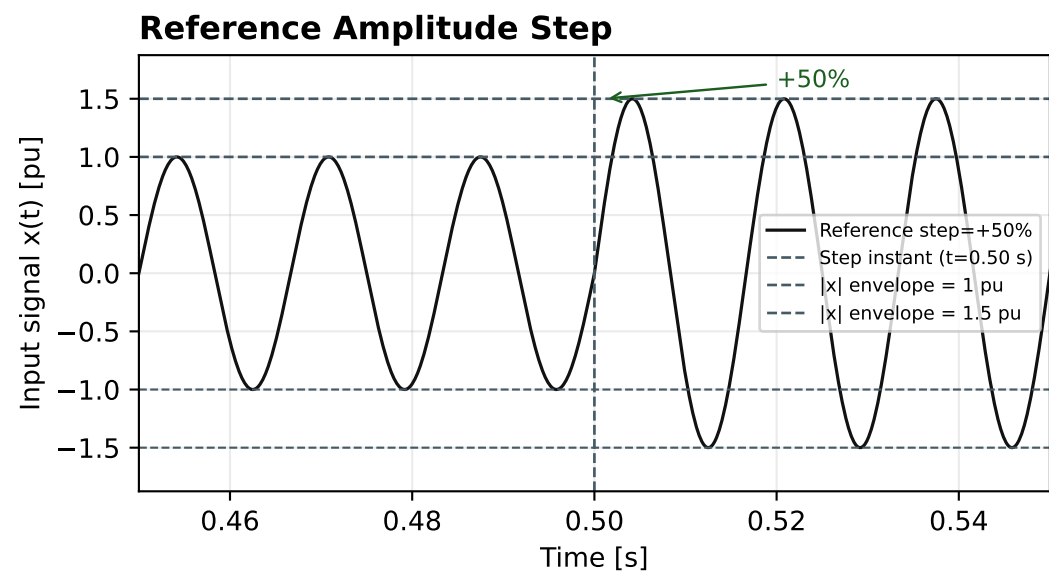


Adaptive



FE max per test: by estimator family

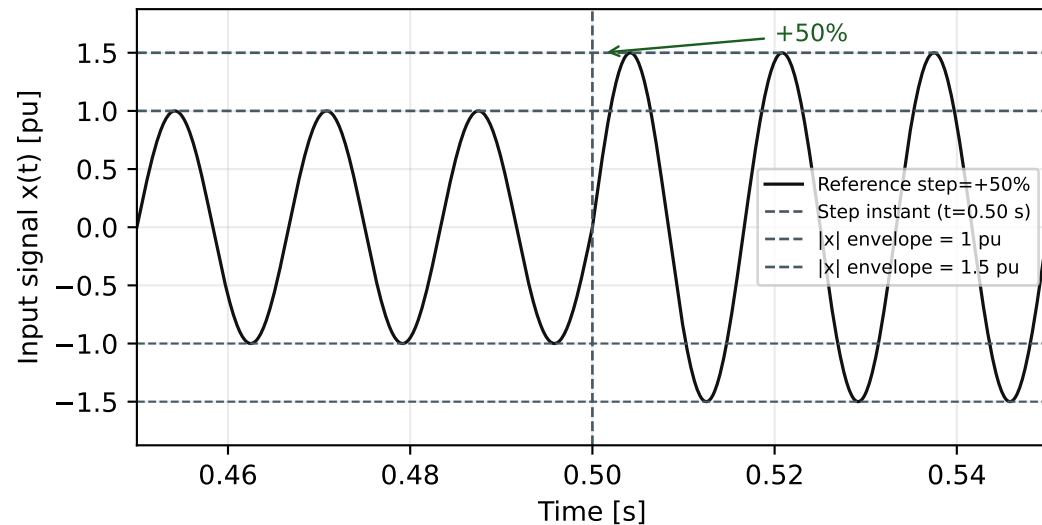
This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify AM-to-FM coupling, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits.



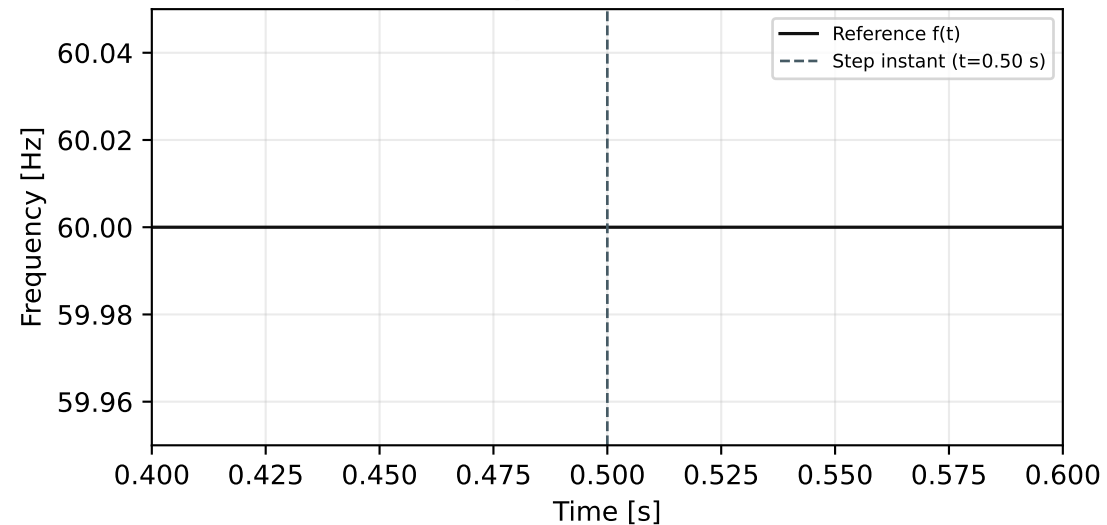
RFE: by estimator family

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify AM-to-FM coupling, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits.

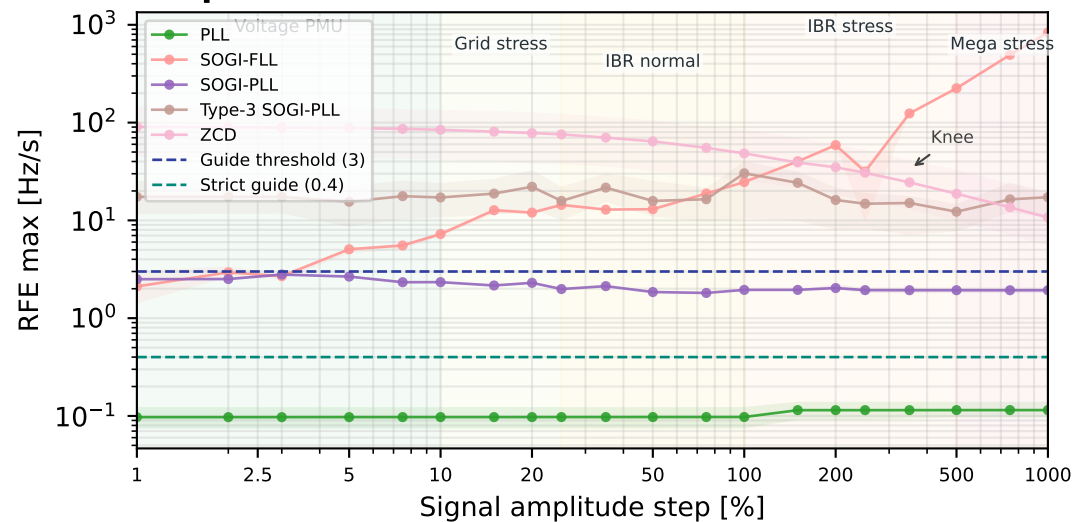
Reference Amplitude Step



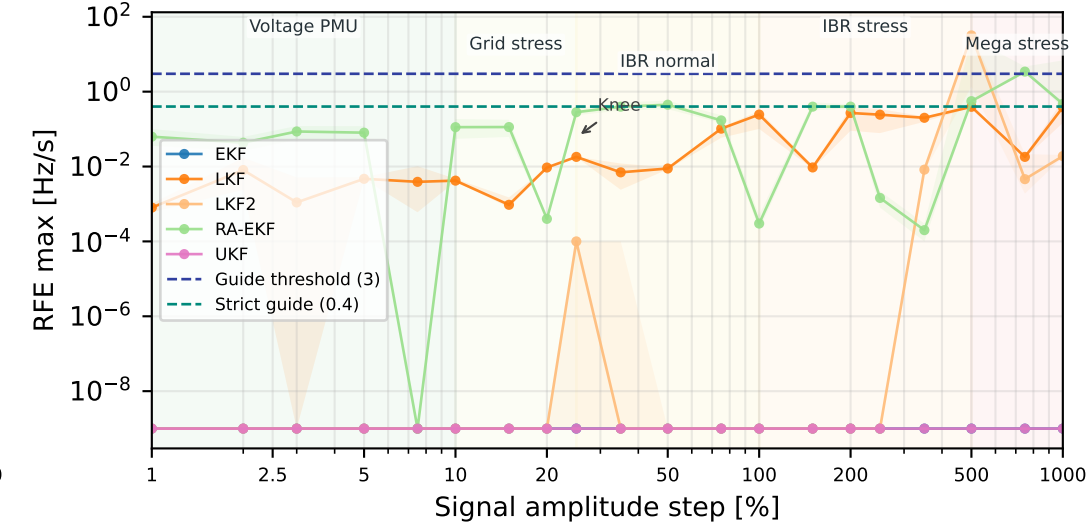
Reference Frequency f(t)



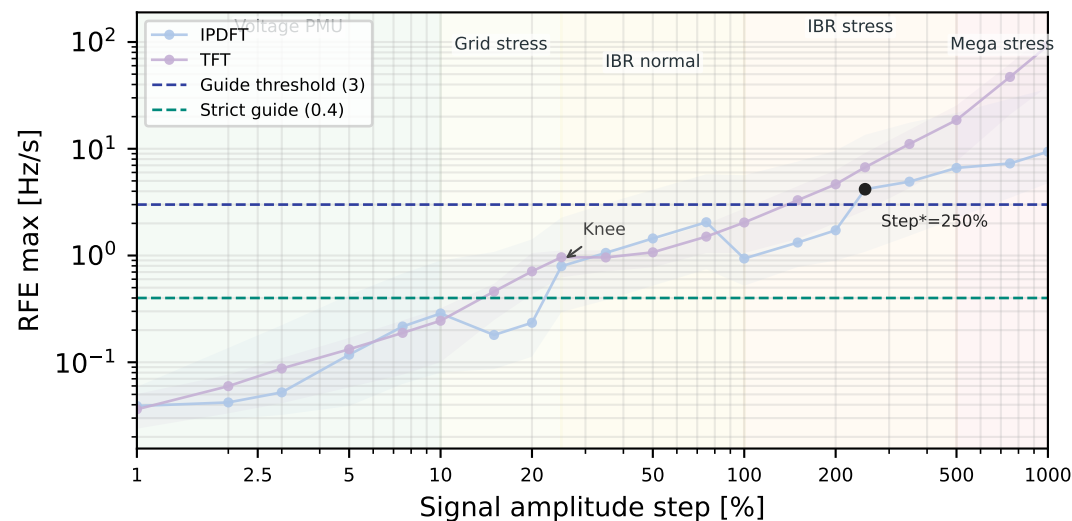
Loop-based



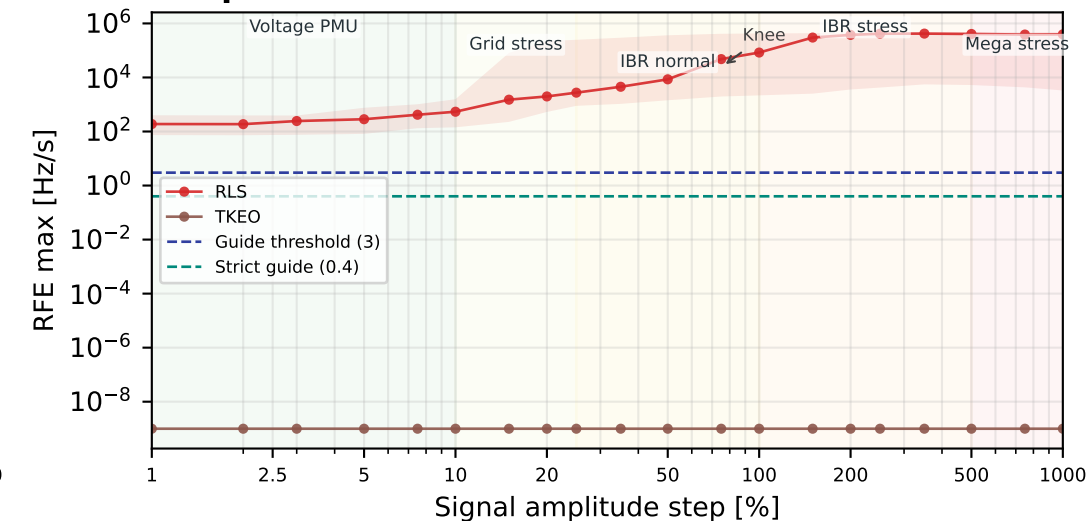
Model-based



Window-based



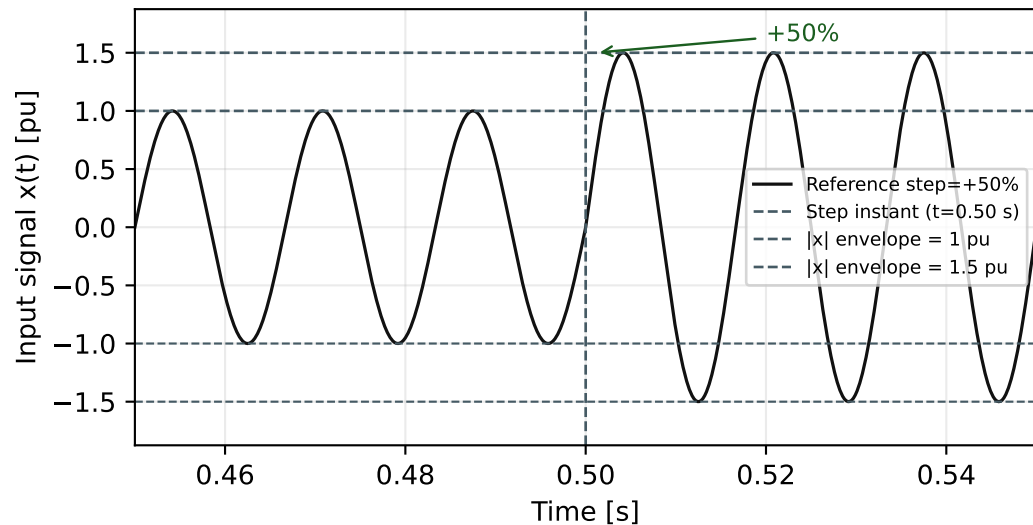
Adaptive



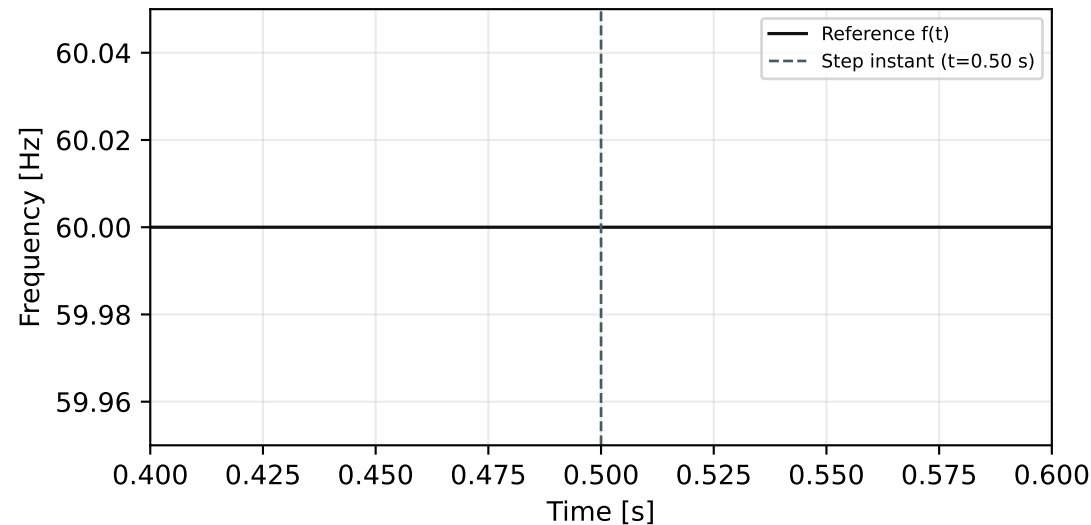
Time out of band: by estimator family

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify AM-to-FM coupling, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits.

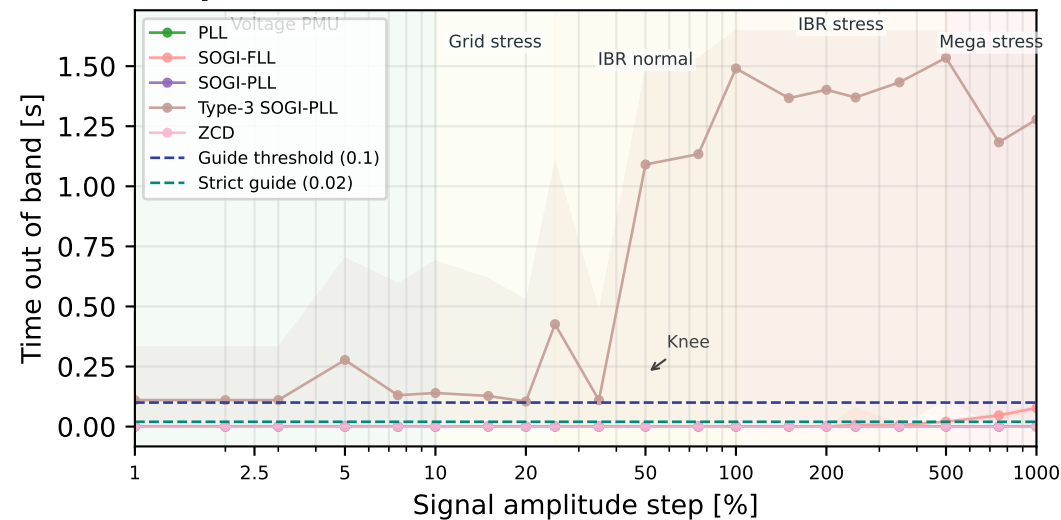
Reference Amplitude Step



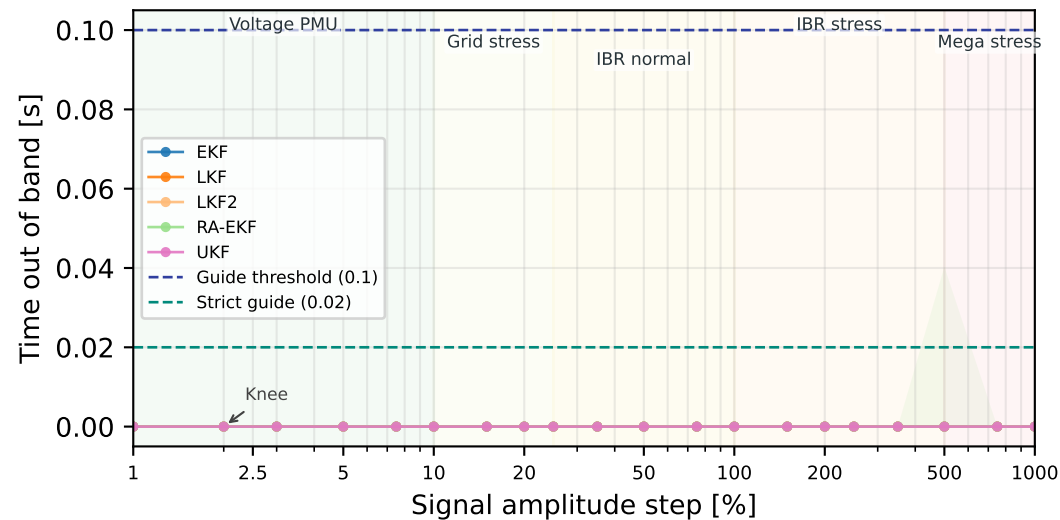
Reference Frequency $f(t)$



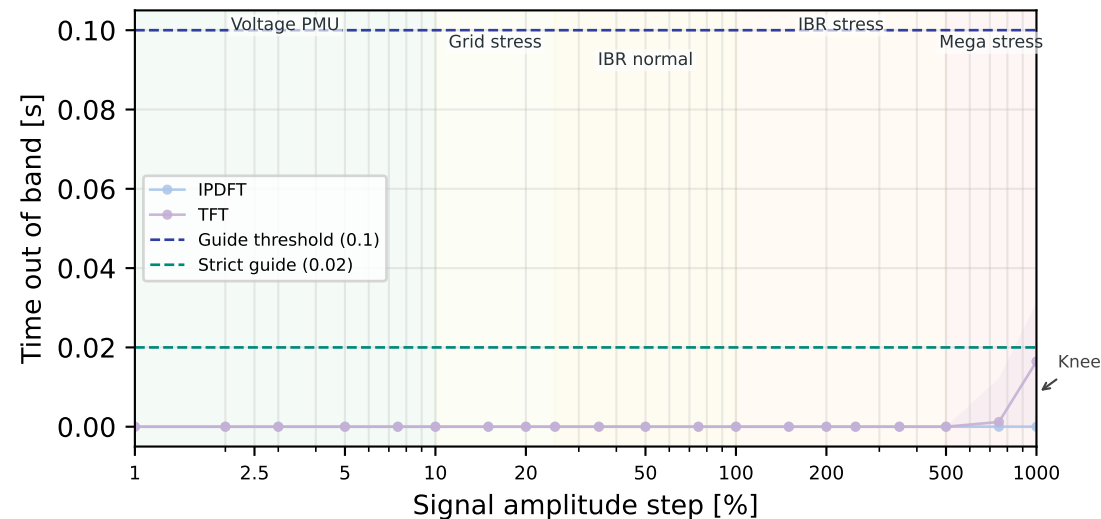
Loop-based



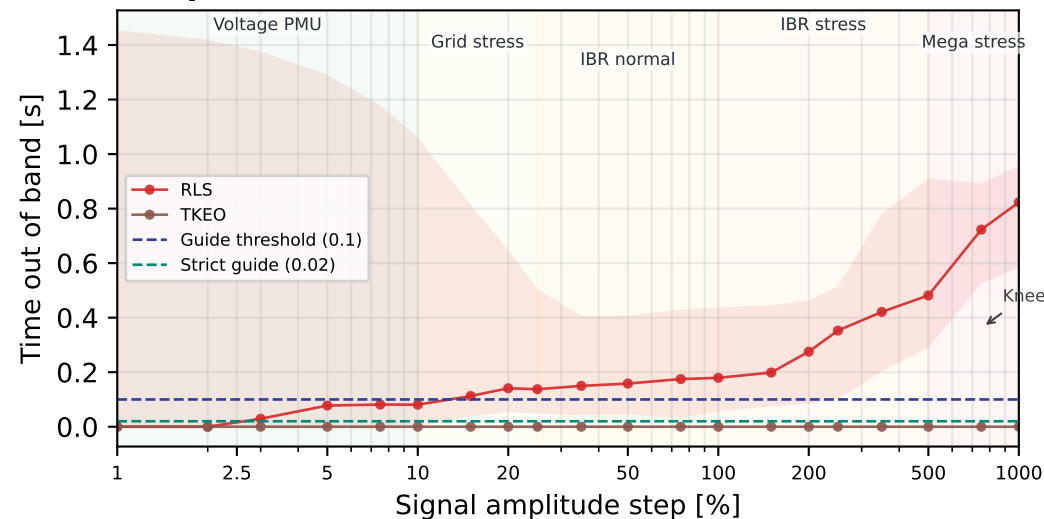
Model-based



Window-based



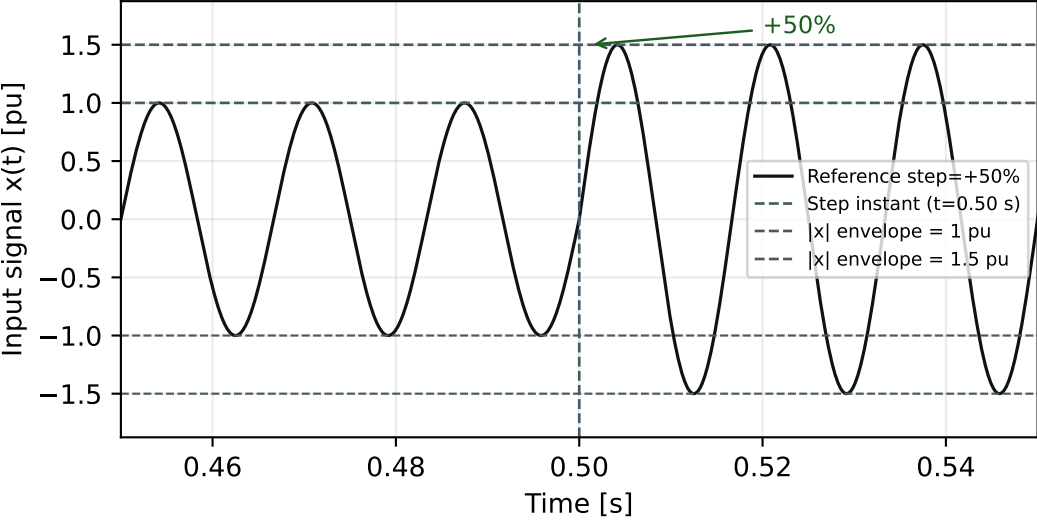
Adaptive



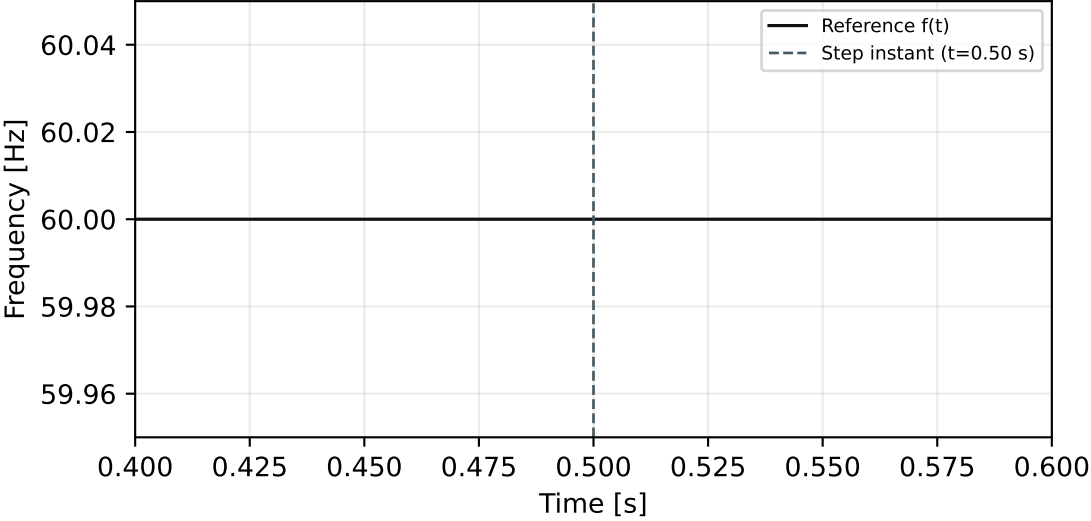
Pass rate: by estimator family

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify AM-to-FM coupling, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits.

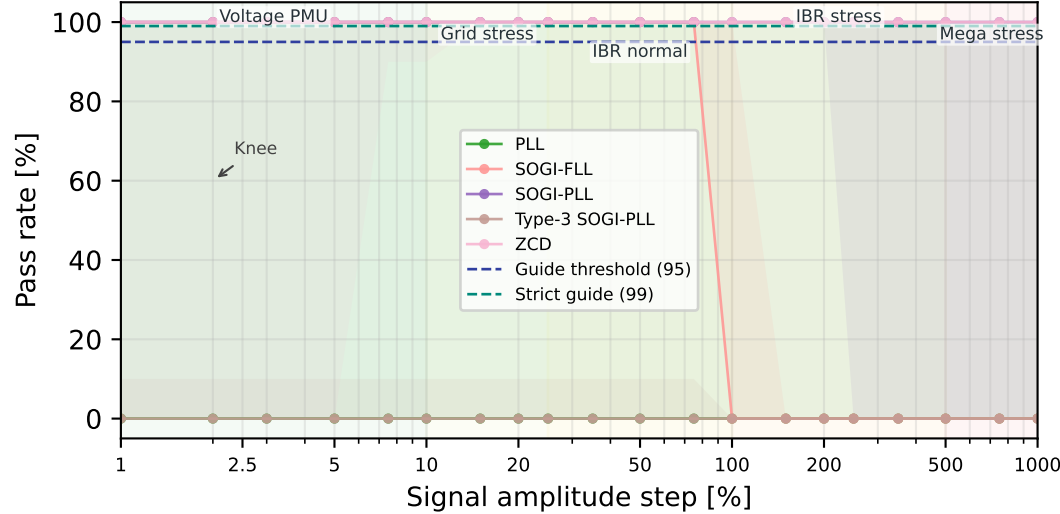
Reference Amplitude Step



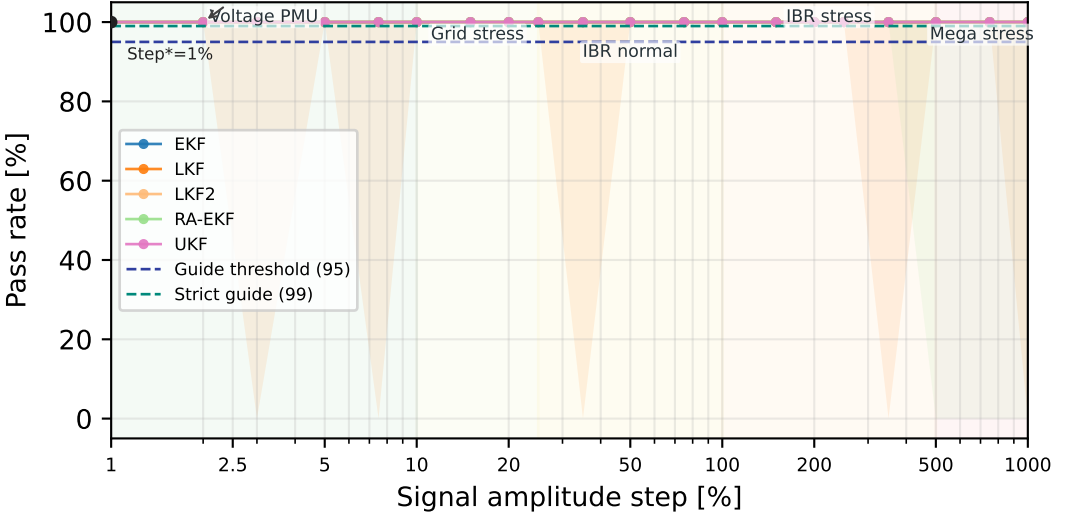
Reference Frequency f(t)



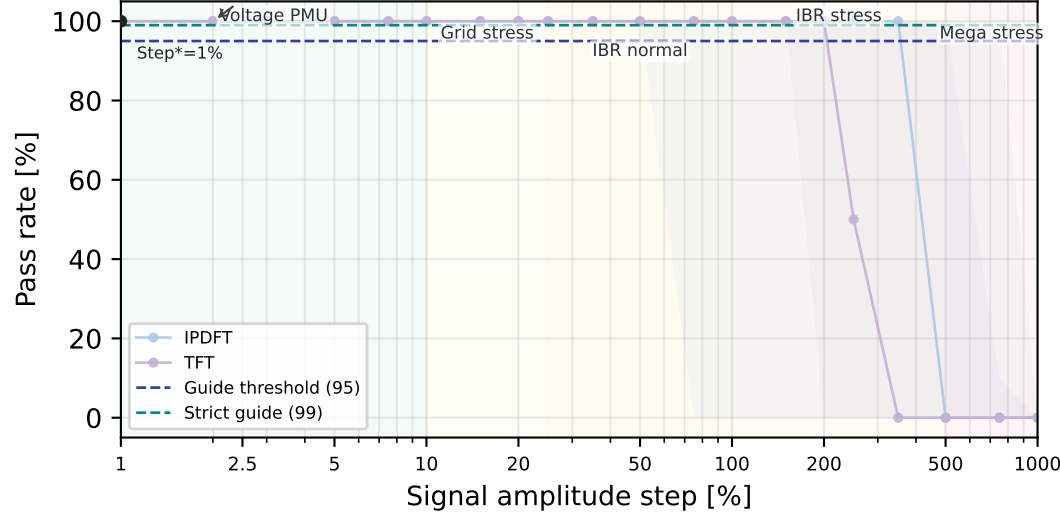
Loop-based



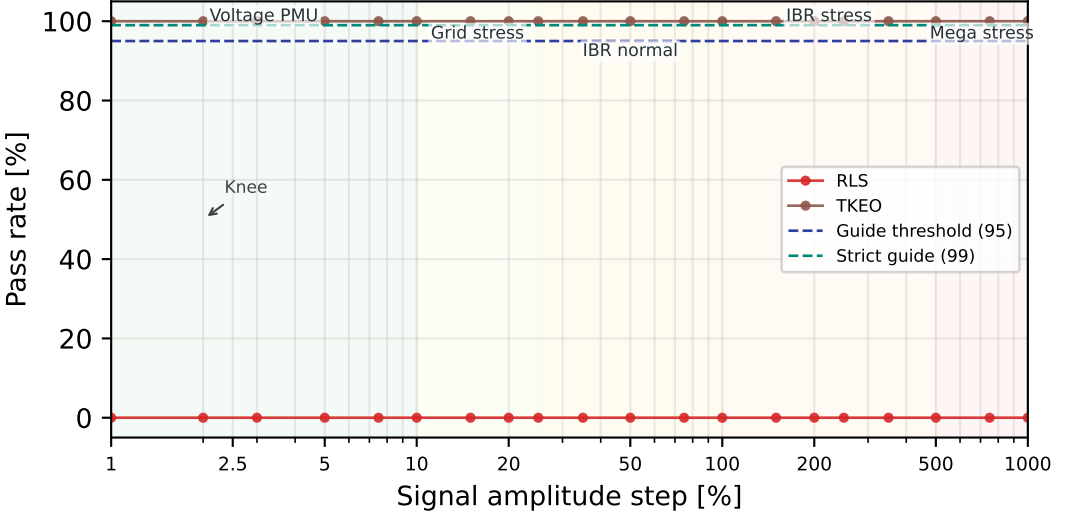
Model-based



Window-based



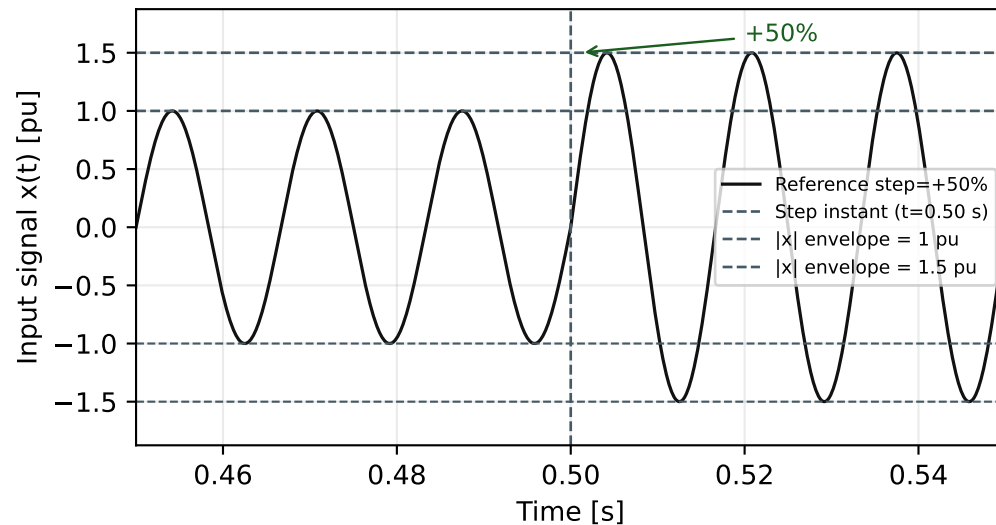
Adaptive



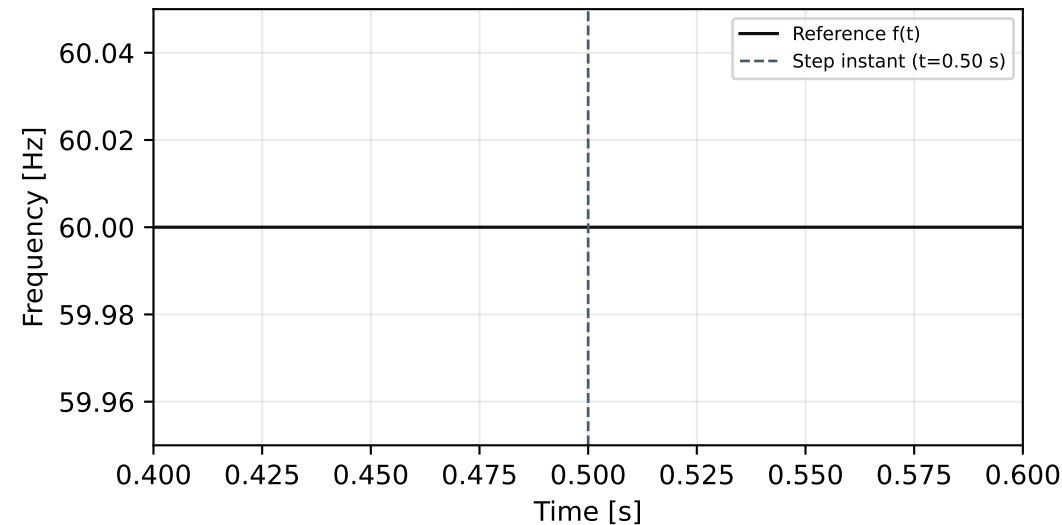
CPU cost: by estimator family

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify AM-to-FM coupling, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits.

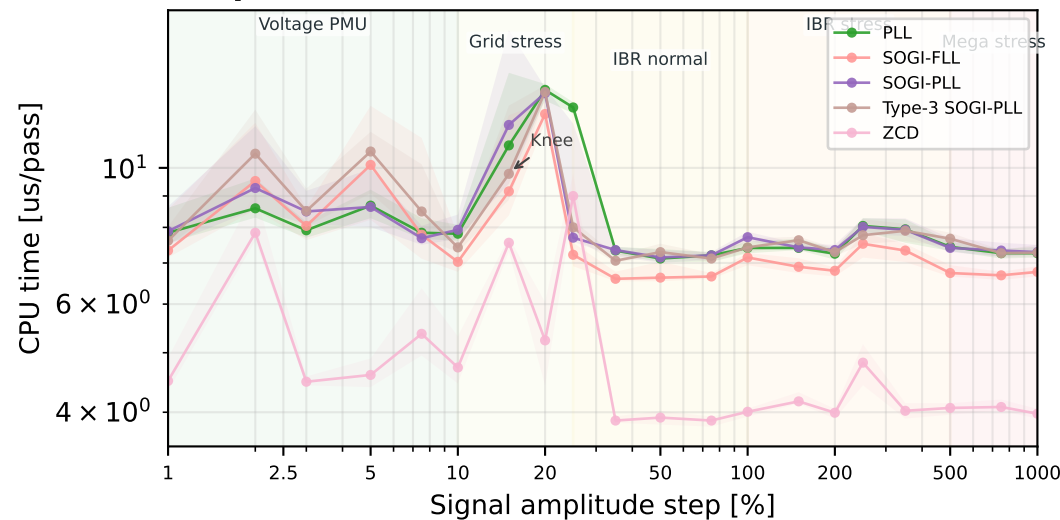
Reference Amplitude Step



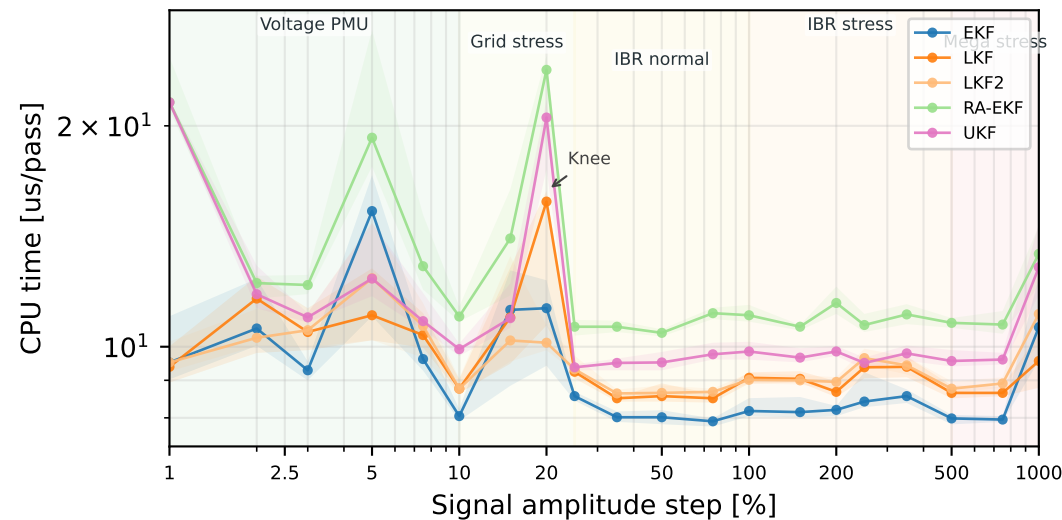
Reference Frequency f(t)



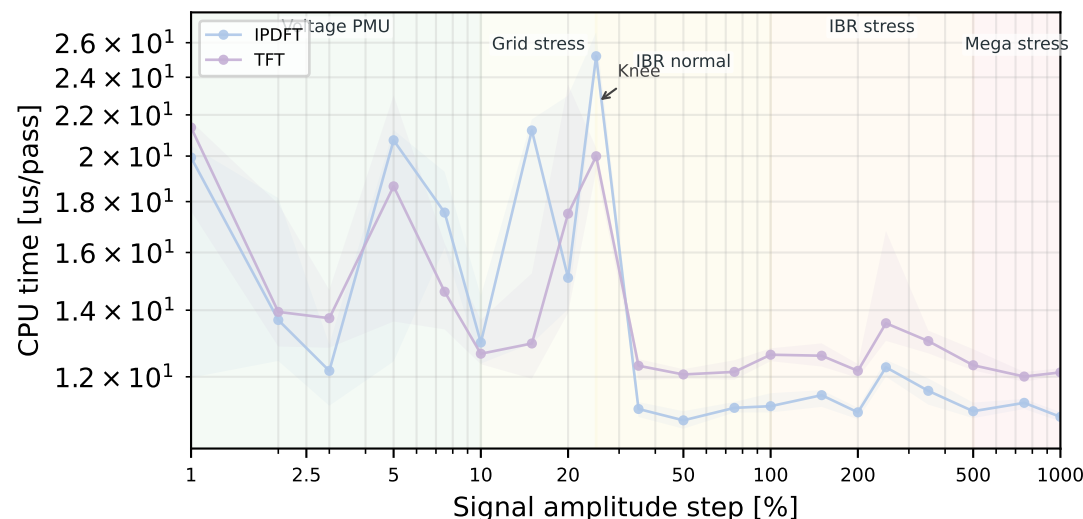
Loop-based



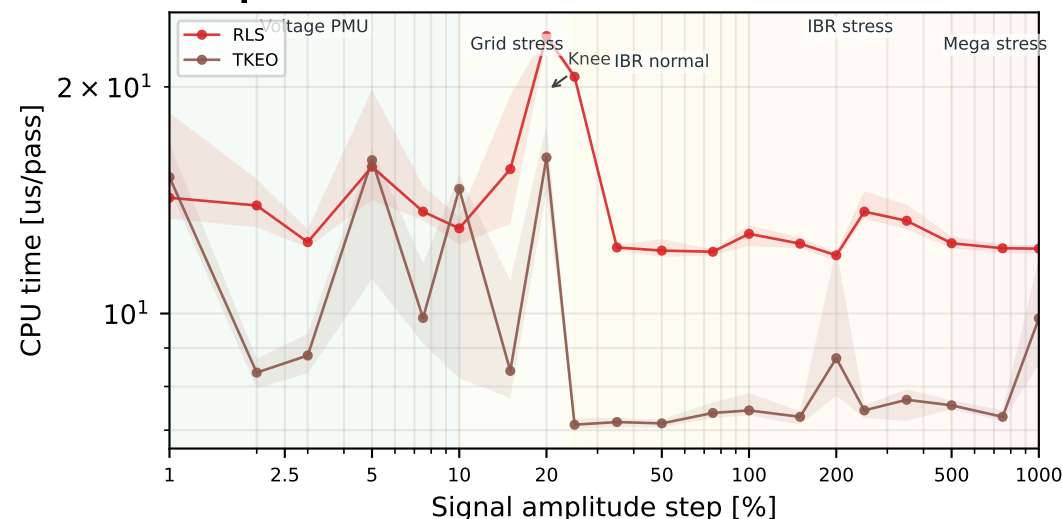
Model-based



Window-based



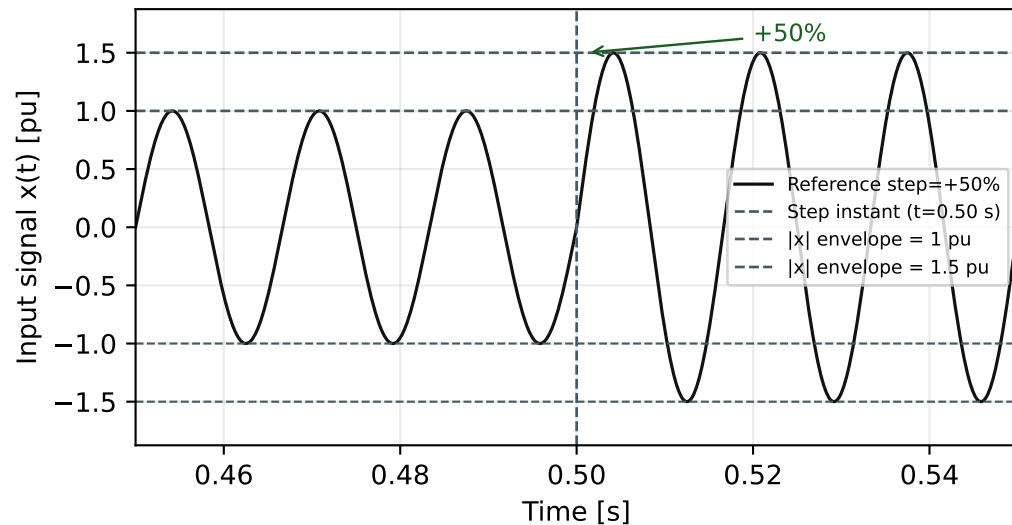
Adaptive



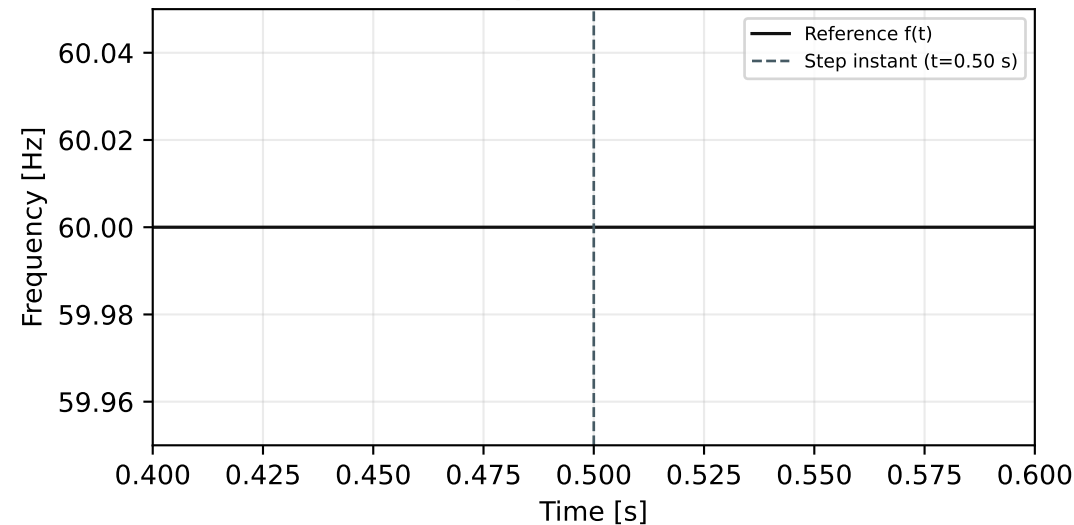
Runtime jitter: by estimator family

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify AM-to-FM coupling, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits.

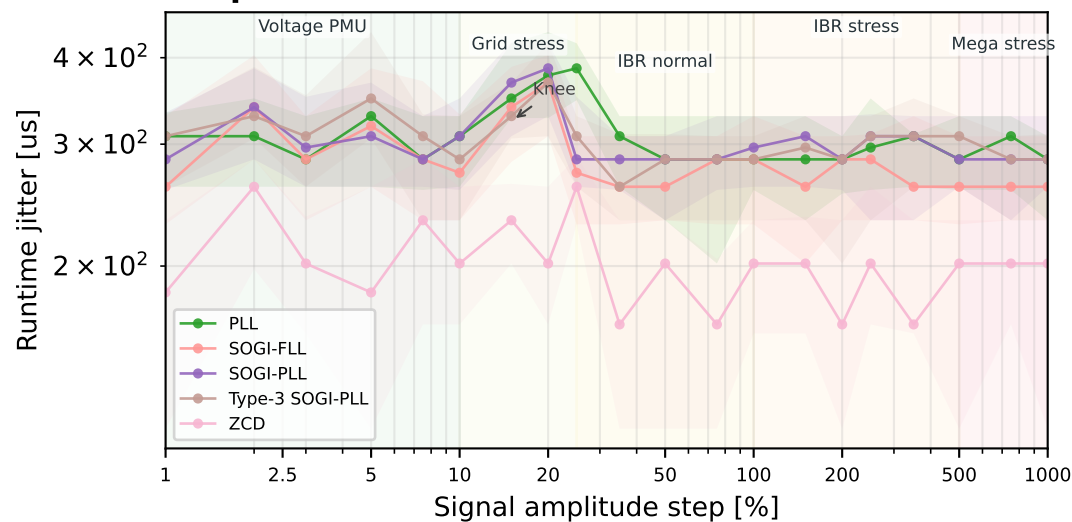
Reference Amplitude Step



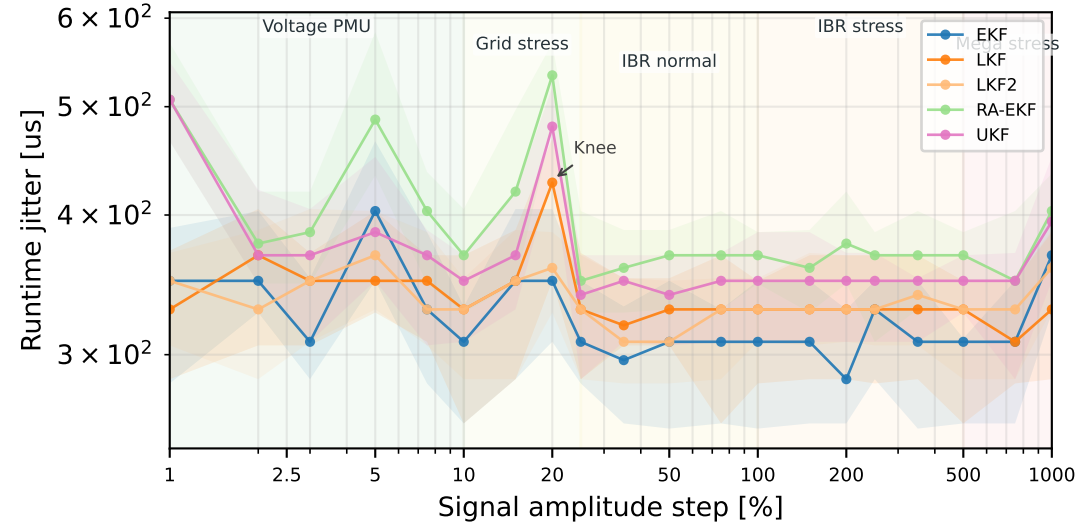
Reference Frequency $f(t)$



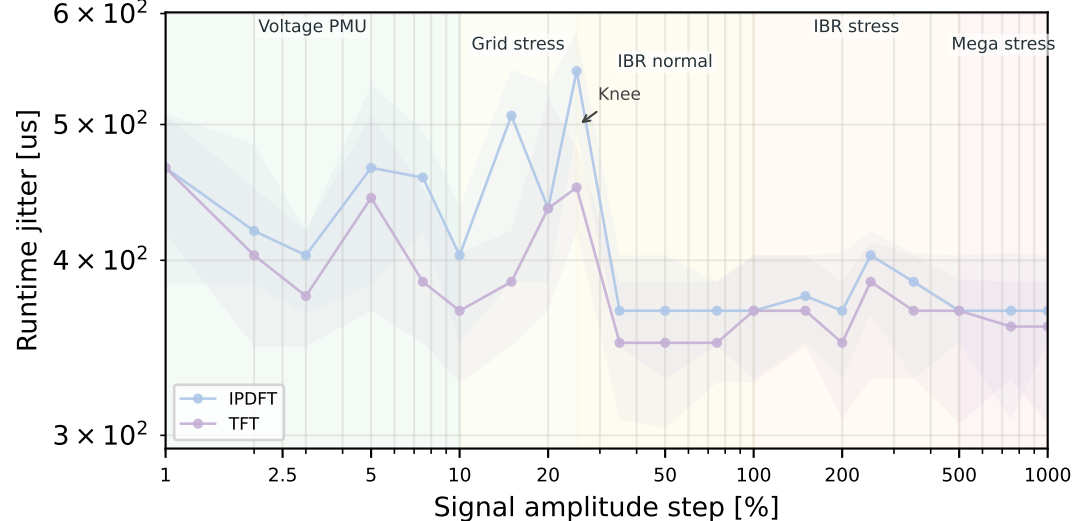
Loop-based



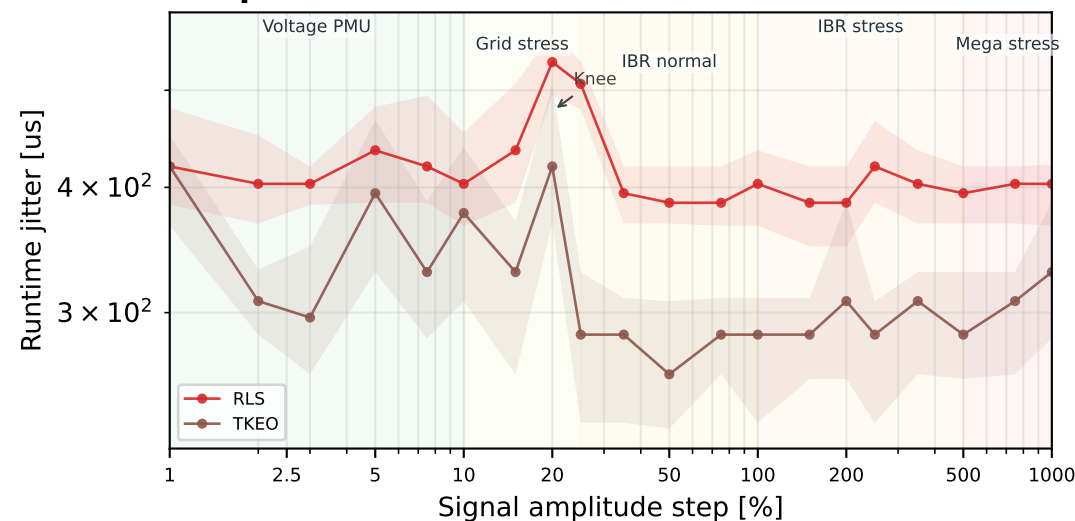
Model-based



Window-based



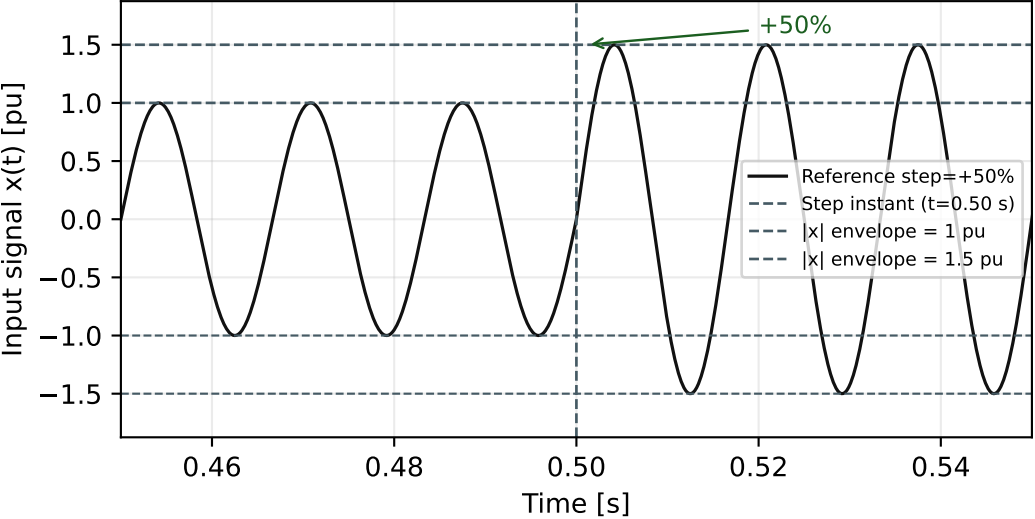
Adaptive



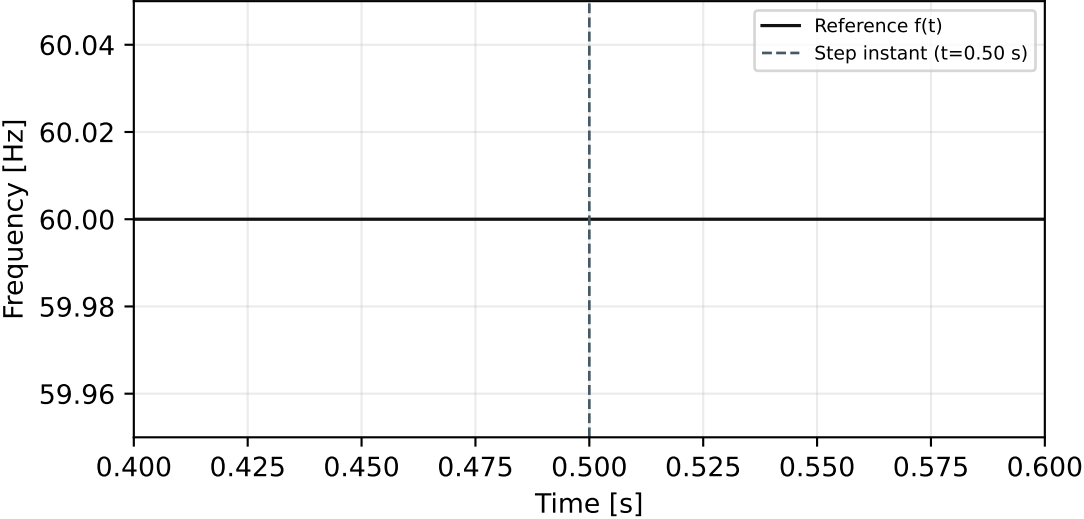
Invalid outputs: by estimator family

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify AM-to-FM coupling, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits.

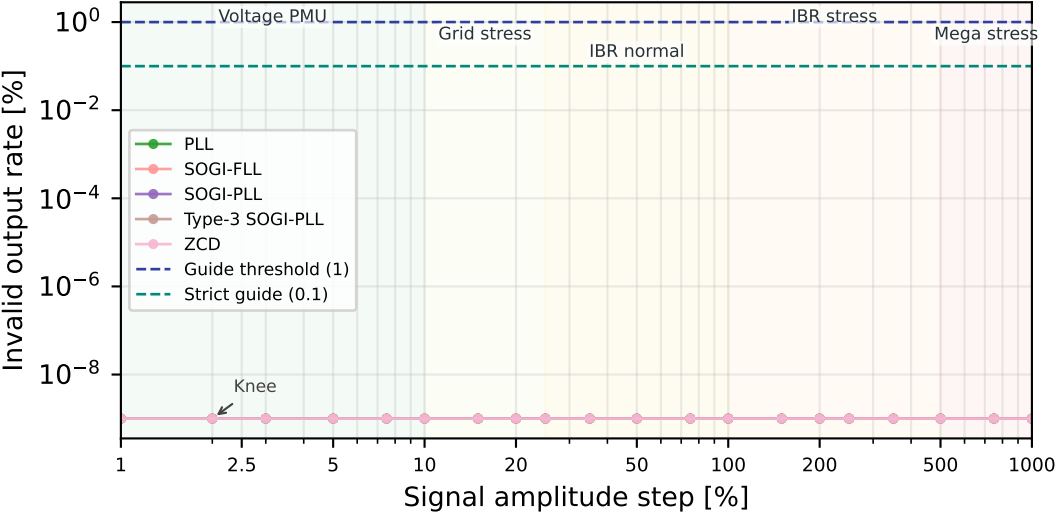
Reference Amplitude Step



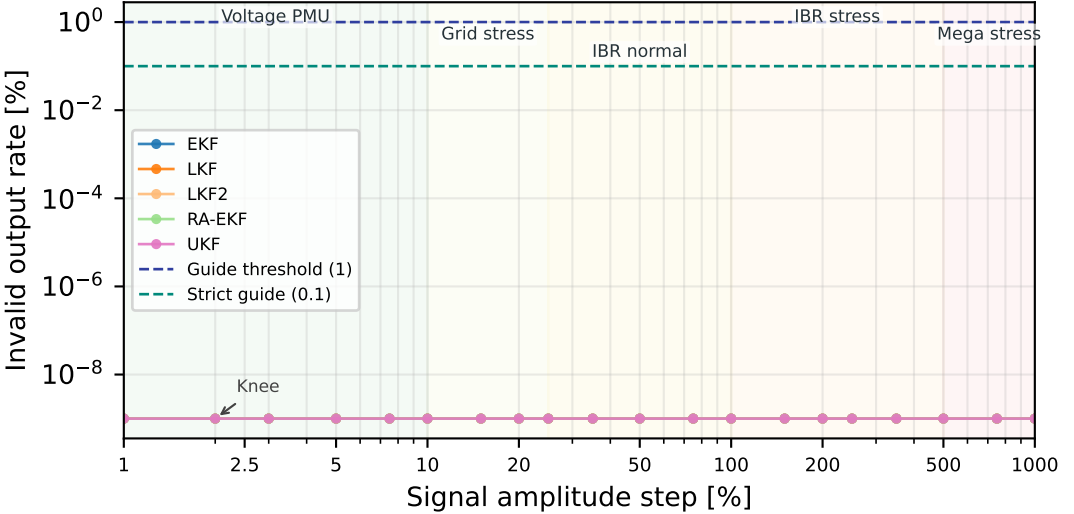
Reference Frequency f(t)



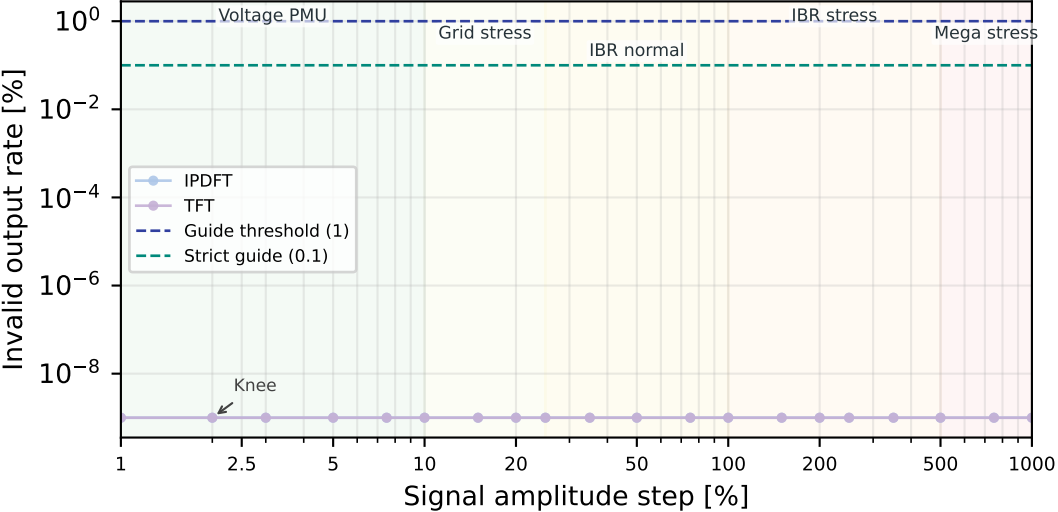
Loop-based



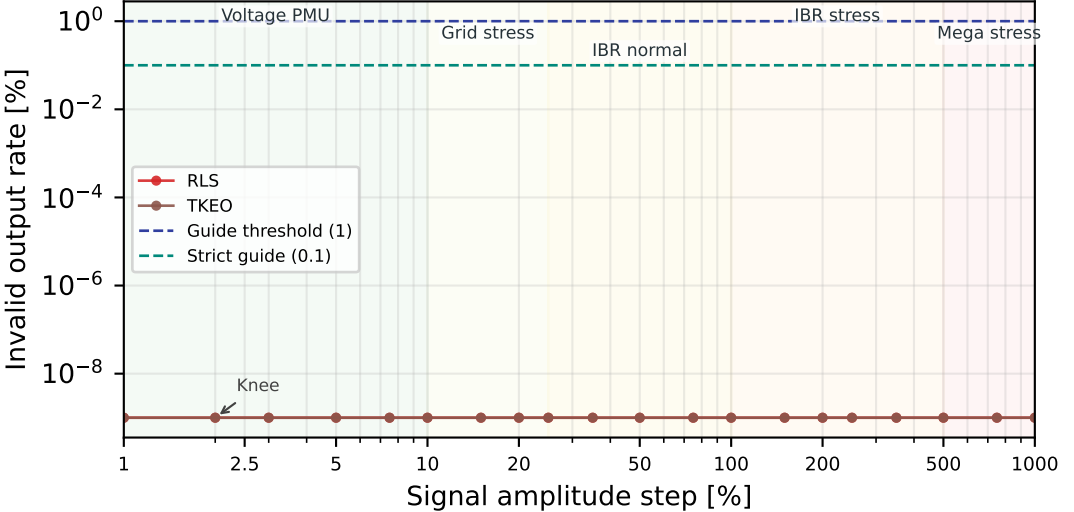
Model-based



Window-based



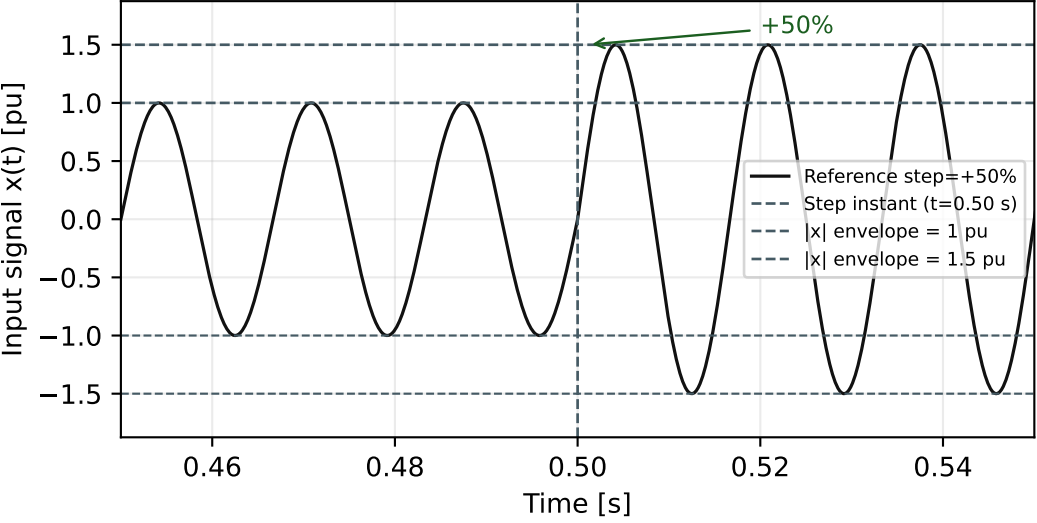
Adaptive



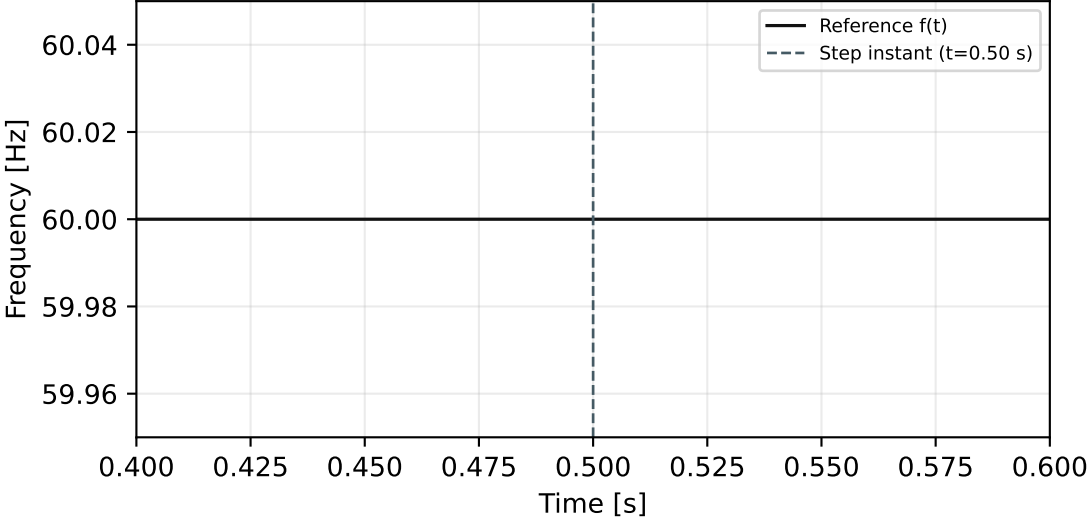
Startup validity: by estimator family

This atlas treats the disturbance as an amplitude-only step: the true frequency remains nominal, so reported frequency and ROCOF errors quantify AM-to-FM coupling, numerical robustness, and tracking stability rather than formal PMU magnitude-step compliance. The 1-1000% schedule combines standard-like voltage perturbations, grid and IBR ride-through stress, and high-amplitude current/fault surrogates; horizontal lines are engineering guide thresholds, not IEEE compliance limits.

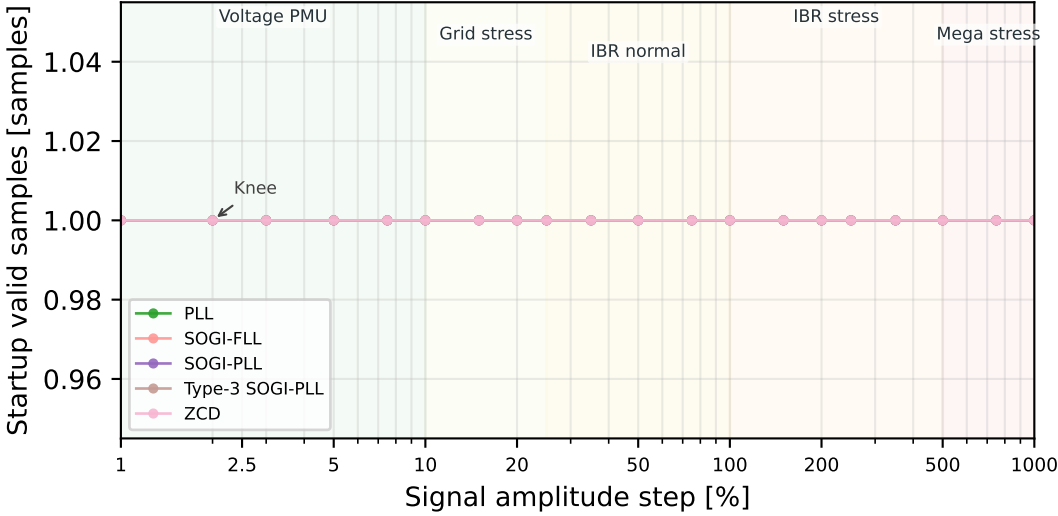
Reference Amplitude Step



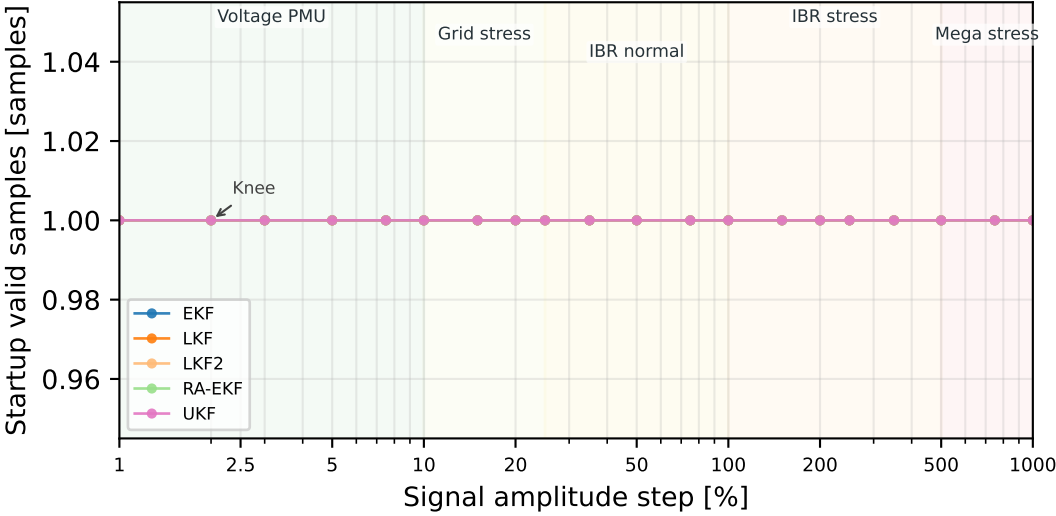
Reference Frequency $f(t)$



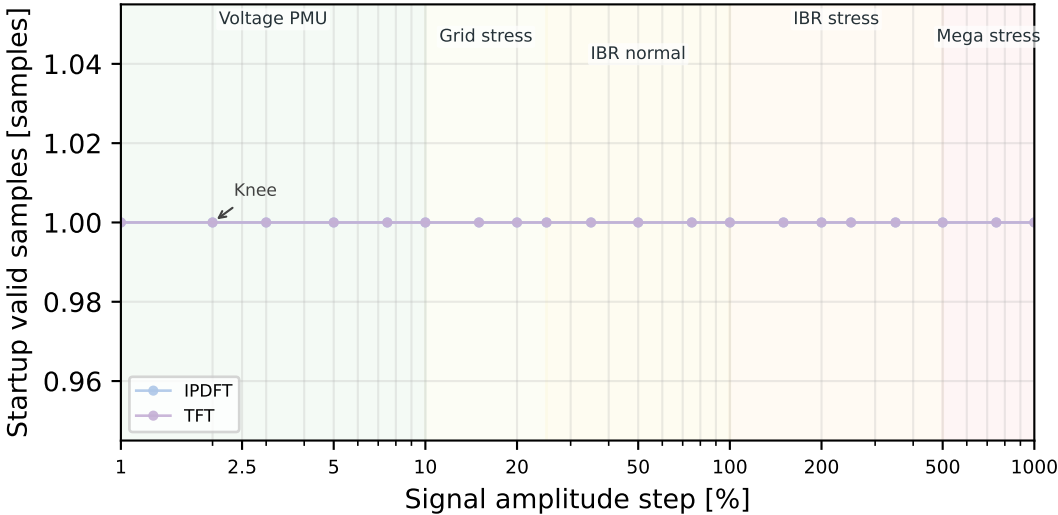
Loop-based



Model-based



Window-based



Adaptive

